Banking Transaction Fraud Analytics

1. Executive Summary

This project analyzed banking-card transaction data to determine where observed fraud exposure is concentrated, which dimensions meaningfully distinguish risk, and where a financial institution could prioritize fraud-monitoring resources.

Metric	Result
Total transactions	13,305,915
Total transaction value	$571.84 million
Fraud-labeled transactions(yes/no)	8,914,963 (67.0%)
Yes fraud-labeled transactions	13,332
fraud rate	0.15% of labeled transactions
Total fraud amount	$1,469,648.78

The strongest finding is that fraud exposure is concentrated more strongly by transaction context—especially merchant category and online channel—than by broad customer demographics. Online transactions account for approximately 72.24% of observed fraud amount, while the six highest-loss merchant categories account for 49.94% of total observed fraud amount.

Merchant categories also reveal different types of risk. Department Stores combine high fraud frequency with the largest aggregate observed loss, whereas Cruise Lines have far fewer fraud events but unusually high loss per fraud transaction. This distinction shows why fraud count, fraud rate, severity and aggregate loss should not be treated as interchangeable.

The business implication is to prioritize targeted monitoring of specific merchant-category × channel combinations and use time/geography as supporting context rather than applying broad controls to entire customer groups or channels.

2. Business Objective

The objective was to identify where historical fraud exposure occurs, how frequently it occurs, how large the associated financial exposure is, and whether risk is disproportionately concentrated within particular transaction segments.

3. Data and Methodology

The project used transaction, customer/user, card, fraud-label and Merchant Category Code (MCC) reference datasets.

The workflow followed opening raw dataset files in python through pandas to identify column names and structure to develop schema for tables in postgresql. Then importing in the database followed by cleaning and validation and identifying correct data type and later forming correct structure as clean tables in the database. Then creating primary/foreign key relationships, consolidated analytics view, independent modules analysis, cross module analysis, advanced sql, business analysis.

Fraud KPIs and fraud-risk analyses were calculated only where fraud_label was available. General transaction analysis could use the complete transaction population, but unlabeled transactions were not assumed to be non-fraud.

Analytical techniques included aggregation, segmentation, conditional aggregation, ranking, cross-dimensional analysis, window functions, Pareto analysis, temporal analysis, month-over-month comparisons, rolling measures and cumulative measures.

- merchant_city = 'ONLINE' was treated as the online channel/location of store.
- Long periods with zero observed fraud were interpreted cautiously because label/temporal coverage can affect the result.

4. Key Business Findings

1. Fraud loss is concentrated in a small set of merchant categories

Evidence: Department Stores account for $225,647.19 (15.35%) of observed fraud amount and Cruise Lines for $185,946.78 (12.65%). The six largest MCCs collectively account for 49.94%.

Interpretation: A relatively small set of MCCs captures a large share of historical exposure, making merchant category useful for prioritization.

2. Online transactions are the strongest broad risk concentration

Evidence: Online transactions contain 8,779 frauds among 1,047,865 labeled transactions (about 0.84%) and $1,061,612.39 in observed fraud amount. Physical transactions contain 4,553 frauds among 7,867,098 labeled transactions (about 0.06%) and $408,036.39.

Interpretation: The online population is much smaller yet accounts for approximately 72.24% of observed fraud amount.

3. MCC × channel(physical/online store) is more informative than either dimension alone

Evidence: Online Cruise Lines show 165 frauds and $185,946.78 in observed fraud amount; online Department Stores show 1,547 frauds and $163,891.28. Department Stores at physical locations show 704 frauds and $61,755.91.

Interpretation: The risk profile changes materially when merchant category and channel are combined. Extreme rates in some combinations require denominator and label-composition caution.

4. Fraud frequency and fraud severity identify different problems

Evidence: Department Stores generated 2,251 fraud transactions and $225,647.19 in observed fraud amount. Cruise Lines generated only 165 fraud transactions but $185,946.78—roughly $1,127 per fraud event versus about $100 for Department Stores.

Interpretation: Department Stores are a high-frequency/high-aggregate-loss problem; Cruise Lines are a high-severity problem.

5. Time becomes more useful when combined with MCC/channel

Evidence: Overall fraud rates peak around 10:00–13:00. Cross-analysis shows important exposure within categories such as Department Stores and Cruise Lines during these hours.

Interpretation: Hour alone is a weaker signal than MCC × hour or channel × hour.

6. Customer age shows modest differentiation

Evidence: Observed fraud rates range from about 0.12% for younger groups to 0.17% for 60+, while the 46–60 and 60+ groups contribute the largest aggregate observed fraud amounts. The 60+ segment deserves attention.

It has:

- the highest fraud rate and fraud count
- 2 fraud counts more than 46–60 segment(highest fraud amount lost) although nearly 5lakh less labeled transactions.
- the second-highest fraud amount

Interpretation: This could indicate that older customers are more vulnerable or have different transaction behaviour. Not claiming why this happens because the dataset doesn't tell the cause.

7. Income demonstrates the difference between rate and aggregate exposure

Evidence: The <25K group has the highest observed rate at 0.20%, while the 25K–50K group has the largest aggregate observed fraud amount at $908,105 and 8,253 fraud transactions.

Interpretation: The highest-rate segment is not the highest-loss segment.

8. Geography contains investigative signals but requires caution

Evidence: Tuvalu, Haiti, Italy are high-risk but probably tiny-volume segments

Top rows are:

- Tuvalu: 5 frauds out of 5 labeled = 100%
- Haiti: 253 frauds out of 264 labeled = 95.83%
- Italy: 3,061 frauds out of 4,706 labeled = 65.04%

Interpretation: These are striking, but they are likely small-sample, high-risk foreign-location segments. They are useful for pattern detection, but not for broad business conclusions unless volumes are large enough.

5. Risk Concentration

Risk type	Primary example	Business meaning
High frequency	Department Stores — 2,251 fraud transactions	Large number of fraud events
High rate	Online channel — ~0.84% vs ~0.06% physical	Fraud elevated relative to labeled exposure
High severity	Cruise Lines — ~$1,127 observed amount per fraud event	Large financial exposure per event
High aggregate loss	Department Stores — $225,647.19	Largest total observed MCC exposure

Pareto analysis shows cumulative observed fraud amount of 15.35% for the top MCC, 28.01% for the top two, 35.75% for the top three, 41.28% for the top four, 45.78% for the top five, 49.94% for the top six, and 60.48% for the top ten. This concentration provides a defensible basis for prioritizing investigative and monitoring resources.

6. Prioritized Business Recommendations

1. Add ETL pipeline for automation using Airflow

2. Powerbi dashboard

3. Machine learning fraud detection model.

7. Weak or Non-Actionable Analyses

Analysis	Observed result	Decision
Gender	Female and Male both ~0.15% observed fraud rate	No meaningful rate discrimination; do not prioritize.
Credit score	Most segments cluster around 0.14–0.15%; Poor is 0.12%	Weak fraud segmentation; creditworthiness and transaction fraud are distinct risks.
Card brand	Mastercard/Visa ~0.15%, Amex 0.16%, Discover 0.21% with smaller exposure	Insufficient evidence for strong brand-specific conclusions.
Card on dark web	All analyzed records = No	No comparison group; non-discriminatory.
Has chip	Chip 0.15%, non-chip 0.13%	Small difference; card capability is not proof of transaction authentication method.

Stopping low-value exploration is an analytical decision. These variables were tested but were not forced into unsupported business conclusions.

8. Limitations

Incomplete fraud-label coverage: Only 67% of transactions are labeled. Reported fraud rates therefore describe the labeled population and may be biased if label availability is non-random.

Temporal gaps: Extended periods with zero observed fraud may reflect data or label coverage rather than true elimination of fraud.

Small-base effects: Very high percentages in small segment combinations can be unstable and should not be generalized.

No fraud-model predictions: False Positive Rate, precision, recall/Fraud Detection Rate and ROC-AUC cannot be calculated without model predictions.

No chargeback outcome: A genuine chargeback rate cannot be calculated without confirmed chargeback lifecycle data.

Fraud mechanism: Aggregated transaction patterns cannot identify whether fraud arose from stolen credentials, account takeover, card-not-present fraud, first-party fraud or another mechanism.

Extreme Percentages: Can result from large small denominators.